CVD. High levels of triglyceride combined with high levels of LDL cholesterol speed up atherosclerosis.

4. **Body weight:** Obesity results when **energy intake** (i.e. food intake) is more than the **energy output** (i.e. physical activity). Obesity often occurs in combination with elevated blood cholesterol level, hypertension, and diabetes thereby increasing the risk of stroke, heart attack and heart failure. Increasing physical activity can help to reduce the risk of obesity and associated CVD.

5. **Unbalanced diet:** A regular intake of foods rich in saturated fat, trans fat, and salt, coupled with a low intake of fruits, vegetables, legumes, whole grains and lean protein is unhealthy and can significantly increase the risk of CVD. A balanced diet which contains foods from different food groups in proportions needed by the body is recommended to reduce risk.

6. **Unhealthy fat:** Some sources of unhealthy fat include red meat, bakery products, butter, junk food, fried foods and pre-packaged snack foods. It is advisable to replace such fat with healthier sources such as vegetable and plant - based oils.

7. **Salt intake:** Increased salt consumption results in increased blood pressure, which contributes to the overall CVD risk. Indians have been consuming far more salt as against the 5 gms per day limit recommended by the World Health Organisation (WHO, 2012).

8. **Dietary fibre:** Dietary fibre is available only from plant sources such as fruits, vegetables, beans, pulses and whole grains. Greater dietary fibre intake is associated with a lower risk of CVD. Research shows that soluble fibre reduces LDL cholesterol levels.

9. **Physical activity:** Being physically active helps in maintaining blood pressure, blood glucose and blood lipid levels within normal limits and thereby reduces the risk of obesity and CVD.

10. **Smoking:** Smoking is extremely harmful to the heart and lungs, and it is the main contributing factor for atherosclerosis and cancer. Passive smoking is also responsible for CVD, and therefore those who smoke are also putting people around them at an increased risk of developing CVD.

11. **Stress :** Stress may lead to an increase in blood pressure and blood lipid levels which may, in turn, increase the risk for CVD. Some individuals try to cope with stress in wrong ways like smoking, consuming alcohol and indulging in comfort eating like consuming high sugar, high fat foods which may result in obesity further increasing their risk for CVD. Positive stress coping strategies that could be adopted include regular physical activity, pursuing a hobby, travelling, yoga and ofcourse taking a balanced diet.

11.12 Diet in CVD

A heart healthy diet ensures a calorie balanced diet that is low in cholesterol, saturated fat, sugar, salt and sodium with an adequate supply of omega 3 fatty acids, complex carbohydrate, protein and dietary fibre.

Carbohydrate

Complex carbohydrate from whole grains, fruits and vegetables are recommended. Excess consumption of refined carbohydrates like maida and simple sugars coupled with overweight and obesity can lead to elevated LDL and triglyceride levels and lower HDL levels.

Dietary fibre

Increased consumption of insoluble as well as soluble dietary fibre reduces the risk of developing CVD by lowering serum LDL cholesterol levels. Besides this, dietary fibre binds with bile acids and makes them unavailable for fat digestion.

UNIT - 11.indd 184 25-02-2019 15:38:21